Session 5. Measures of centrality.

Pre course: Statistics.

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Agenda

- Centrality measures.
- Exercise 1.
- Impact of data transformations on centrality measures.
- Centrality measures in a histogram.
- Exercises 2 and 3.

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Centrality measures.

- Give a single representative value. Help to compare groups.

- Mean
  - The deviations from the mean always add up zero.
- Median
  - With an odd/even number of observations.
- Mode
  - The most common value of the sample or population

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \ldots + x_n}{n} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

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Exercise 1.

- See excel file, sheet “E5. Exercise 1”.

EXERCISE 1. MEASURES OF CENTRALITY AND OUTLIERS.

4	3	10	3	3	0	3	0	7	30

a) Calculate the mean, median and mode of the 10 numbers given.
b) Calculate again the mean, median and mode but this time remove the outlier (i.e. the number 30).
c) Which measures are more sensitive to outliers, and which are less sensitive to them?

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Impact of data transformations on centrality measures.

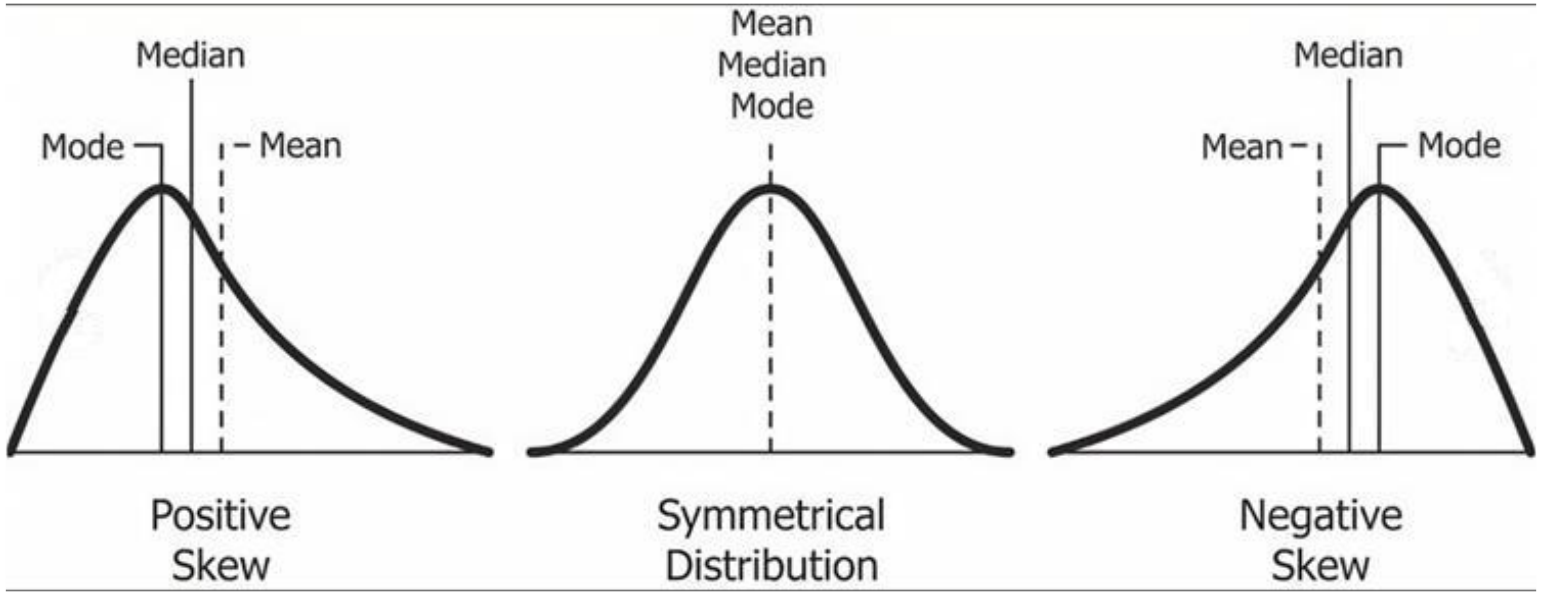
- See excel file, sheet “S5. Transf. centrality measures”.
- Addition.
- Multiplication.
- Outlier, huge outlier and not outlier.

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Centrality measures in a histogram.

- See excel file, sheet “S5. Transf. centrality measures”.



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Exercise 2.

EXERCISE 2. EQUATIONS.

a) The following set of numbers has a mean of 5. Find the value of X.

[2, 4, 8, 2, 9, 10, X]

b) The following set of numbers has a mean of 6. Find the value of X.

[4, 6, 11, 4, 3, 2, X]

c) The following set of numbers has a mean of 7. Find the value of X.

[5, 12, 8, 6, 3, 7, X]

d) The following set of numbers has a mean of 8. Find the value of X.

[5, 2, 9, 12, 10, 11, X]

Exercise 3.

EXERCISE 3. BASIC BASKET.

Context:

1. *A recent study by the statistics office of a middle-income country estimated the cost of a basic basket that includes food, housing, education, and leisure. This estimation reflects the needs of the average person and is methodologically right. The per capita cost is approximately 500 USD.*
2. *A parliamentarian from the governing party claims that families are doing very well because the national average per capita income is around 600 USD.*
3. *When you look at the data in more detail, you see that the average amount mentioned by the parliament member is correct. However, you also find that the median income per person is about 300 USD.*

Question

a) What would you say about the statement made by the parliament member regarding the population's ability to cover the costs of the basic basket?